

AVL Tree Rotations & Operations -- Data Structures

Definition: An AVL Tree (Adelson-Velsky and Landis) is a self-balancing Binary Search Tree in which the Balance Factor (BF) of every node is either -1, 0, or +1.

1. Detailed Technical Explanation

Balance Factor Formula:

Balance Factor (BF) = Height(Left Subtree) - Height(Right Subtree)

Valid AVL Condition: BF $\in \{-1, 0, +1\}$

If after an insertion or deletion, the balance factor of any node becomes ≤ -2 or $\geq +2$, the tree is unbalanced and must be rebalanced using Rotations.

2. The Four AVL Tree Rotations

1. LL Rotation (Single Right Rotation)

* Cause: Insertion into the Left subtree of the Left child of node z (BF of z = +2, BF of left child = +1).

```

      [ z ] (+2)           [ y ] (0)
      /                   /  \
     [ y ] (+1)          [ x ]  [ z ]
      /
     [ x ]

=== Right ==>
Rotation at z
  
```

2. RR Rotation (Single Left Rotation)

* Cause: Insertion into the Right subtree of the Right child of node z (BF of z = -2, BF of right child = -1).

```

      [ z ] (-2)           [ y ] (0)
      \                   /  \
     [ y ] (-1)          [ z ]  [ x ]
      \
     [ x ]

=== Left ==>
Rotation at z
  
```

3. LR Rotation (Double Rotation: Left then Right)

* Cause: Insertion into the Right subtree of the Left child of node z (BF of z = +2, BF of left child = -1).

```

      [ z ] (+2)           [ z ] (+2)           [ x ] (0)
      /                   /                   /  \
     [ y ] (-1)          [ x ]              [ y ]  [ z ]
      \      == Left => /                   == Right => /  \
     [ x ]      on y  [ y ]              on z          [ y ]  [ z ]
  
```

4. RL Rotation (Double Rotation: Right then Left)

* Cause: Insertion into the Left subtree of the Right child of node z (BF of z = -2, BF of right child = +1).

3. Time Complexity of AVL Tree Operations

Operation	Average Case	Worst Case
Search	$O(\log N)$	$O(\log N)$
Insertion	$O(\log N)$ (at most 2 rotations)	$O(\log N)$

Deletion	$O(\log N)$ (up to $O(\log N)$ rotations)	$O(\log N)$
--------------	---	-------------

4. Must-Write Points for Exams

- * The height of an AVL tree with N nodes is strictly bounded by $1.44 * \log_2(N)$.

* AV

5. Quick Recall Flow

Balance Factor = $H(\text{Left}) - H(\text{Right})$ $\{-1, 0, 1\}$ -> Imbalance: LL (Right Rotate), RR (Left Rotate), LR (Left-Right), RL (Right-Left)
--